Phosphate buffer system in kidney

After the absorption of available bicarbonate ions in tubular filtrate, remaining hydrogen ions interact with HPO_4^{2-} or $NaHPO_4^-$ and form $H_2PO_4^-$ or NaH_2PO_4 which can be excreted in urine. Thus, hydrogen ions are removed from extracellular fluid.

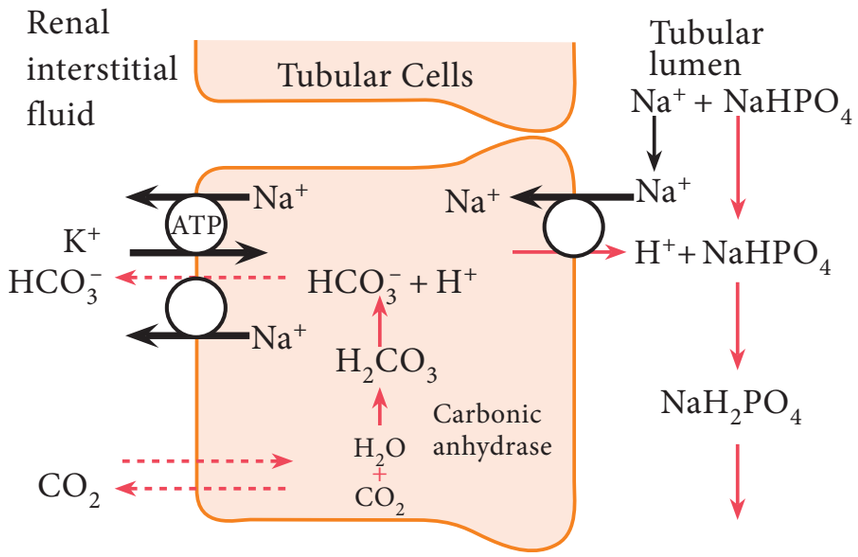

Figure 1.23 Role of phosphate ions in kidney tubule

Ammonium buffer system in kidney

Ammonia produced from glutamine in tubular epithelial cells, can freely diffuse into tubular lumen. This ammonia then combines with hydrogen ions to form ammonium ions which can be easily excreted in urine. Thus, hydrogen ions are removed from extracellular fluid.

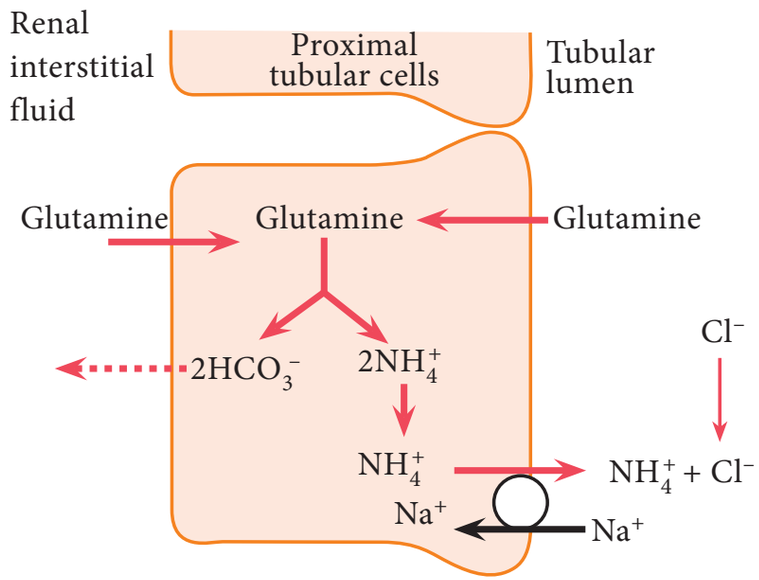

Figure 1.24 Role of ammonium in kidney tubule

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